

ISHITA KAKKAR

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EDUCATION

- University of Wisconsin-Madison (UW-Madison)** Sep 2025 - May 2030 (expected)
Doctor of Philosophy in Computer Science, GPA: 4.0
- University of Massachusetts Amherst (UMass Amherst)** Sep 2021 - May 2025
Bachelor of Science in Computer Science (Honors) and Mathematics, Major GPA: 3.87
- Awards:** NSF GRFP Awardee 2025, UMass Outstanding Undergraduate Achievement Award for Research Excellence

RESEARCH EXPERIENCE

- UW-Madison Hulab** Sep 2025 - Present
Graduate Research Assistant, Advisor: Junjie Hu
- Engineered **helper-objective-guided parent selection** for an **agentic recursive self-improvement loop**: a two-level non-stationary bandit that residualizes helper signals against the primary objective while dynamically discovering and pruning an LLM-generated pool of helper objectives.
 - Demonstrated up to $\sim 2\times$ faster convergence to high-quality solutions than a primary-objective-only baseline in early experiments across multiple optimization domains like circle packing and job-shop scheduling (e.g., reaching a strong-solution threshold in ~ 95 vs. ~ 190 generations).
 - First author and research lead on this ongoing project.
- IBM Research, MIT-IBM Watson AI Lab** Aug 2024 - Dec 2024
Research Intern, Advisor: Mikhail Yurochkin
- Evaluated strengths and weaknesses of different-sized LLMs across prompt topics by combining topic modeling (BERTopic, k-means) with human feedback analysis and generated insights that informed LLM query-routing strategies for cost-efficient inference.
- Northeastern University** Sep 2023 - Aug 2024
Research Intern, Advisor: Mai ElSherief
- Developed EXPRESS, a benchmark and evaluation framework for fine-grained emotion recognition in LLMs using Reddit data with extended contexts and multi-metric analysis, uncovering gaps in LLM emotion recognition and alignment with human self-disclosures.
- University of California San Diego** Jun 2023 - Aug 2024
Research Intern, Advisor: Mai ElSherief
- Developed a contextual word sense disambiguation approach and released a 2,300+ post Reddit dataset on mental burnout, training ML models that achieved ≥ 0.85 F1 in classifying mental health relevance and burnout context.

PUBLICATIONS

- [1] **Ishita Kakkar**, Enze Zhang, Rheeja Uppaal, Junjie Hu. “HarmThoughts: A Benchmark for Fine-Grained Harmful Behavior Detection in Reasoning Traces.” *Proceedings of Conference on Language Modeling (COLM) 2026, Main Track*. [Paper]
- [2] Bangzhao Shu, Isha Joshi, Melissa Karnaze, Anh C. Pham, **Ishita Kakkar**, Sindhu Kothe, Arpine Hovasapian, and Mai ElSherief. “Fluent but Unfeeling: The Emotional Blind Spots of Language Models.” *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM), 2026*. [Paper]
- [3] Nazanin Sabri, Anh Pham, **Ishita Kakkar**, and Mai ElSherief. “Inferring Mental Burnout Discourse Across Reddit Communities.” *Proceedings of the Third Workshop on NLP for Positive Impact (NLP4PI), EMNLP 2024, pp. 224–231*. [Paper]

GRADUATE COURSEWORK

AI Agents for Science, Stochastic Computational Methods, Mathematical Foundations of Machine Learning, Research Methods in Empirical Computer Science, Computer Vision, Advanced Deep Learning, Teaching Assistant for Big Data Systems, Teaching Assistant for Advanced Natural Language Processing